

Supplement S1: Exact finite classification tables

This supplement is a literal finite presentation of the already-proved symbolic classification; it is not a premise of the all- k theorem.

P	δ_0	δ_1	τ	ρ	regime	transport	$C_P(k)$
MMMMMM	1	1	1	1	ZERO	N/A	0
MMMMMV	1	-1	1	1	ZERO	N/A	0
MMMMVM	-1	1	1	1	ZERO	N/A	0
MMMMVV	-1	-1	1	1	CONSTANT_RHO_PLUS	I	6k-5
MMVMVM	1	-1	-1	1	ZERO	N/A	0
MMVMVV	1	1	-1	1	ZERO	N/A	0
MMVMVM	-1	-1	-1	1	PLEAT	I	1
MMVMVV	-1	1	-1	1	ZERO	N/A	0
MMVMMM	-1	1	-1	1	ZERO	N/A	0
MMVMMV	-1	-1	-1	1	PLEAT	R	1
MMVMVM	1	1	-1	1	ZERO	N/A	0
MMVMVV	1	-1	-1	1	ZERO	N/A	0
MMVMMM	-1	-1	1	1	CONSTANT_RHO_PLUS	R	6k-5
MMVMVV	-1	1	1	1	ZERO	N/A	0
MMVVVM	1	-1	1	1	ZERO	N/A	0
MMVVVV	1	1	1	1	ZERO	N/A	0
MVMMMM	-1	-1	1	-1	CONSTANT_RHO_MINUS	I	4k-3
MVMMMV	-1	1	1	-1	ZERO	N/A	0
MVMMVM	1	-1	1	-1	ZERO	N/A	0
MVMMVV	1	1	1	-1	ZERO	N/A	0
MVMVMM	-1	1	-1	-1	ZERO	N/A	0
MVMVMV	-1	-1	-1	-1	PLEAT	I	1
MVMVVM	1	1	-1	-1	ZERO	N/A	0
MVMVVV	1	-1	-1	-1	ZERO	N/A	0
MVVMMM	1	-1	-1	-1	ZERO	N/A	0
MVVMMV	1	1	-1	-1	ZERO	N/A	0
MVVMMV	-1	-1	-1	-1	PLEAT	CH	1
MVVVMV	-1	1	-1	-1	ZERO	N/A	0
MVVVMM	1	1	1	-1	ZERO	N/A	0
MVVVMV	1	-1	1	-1	ZERO	N/A	0
MVVVVV	-1	1	1	-1	ZERO	N/A	0
MVVVVV	-1	-1	1	-1	CONSTANT_RHO_MINUS	CH	4k-3
VMMMMM	-1	-1	1	-1	CONSTANT_RHO_MINUS	H	4k-3
VMMMMV	-1	1	1	-1	ZERO	N/A	0
VMMMVM	1	-1	1	-1	ZERO	N/A	0
VMMMVM	1	1	1	-1	ZERO	N/A	0
VMMVMM	-1	1	-1	-1	ZERO	N/A	0
VMMVMV	-1	-1	-1	-1	PLEAT	H	1
VMMVVM	1	1	-1	-1	ZERO	N/A	0
VMMVVV	1	-1	-1	-1	ZERO	N/A	0
VMVMMM	1	-1	-1	-1	ZERO	N/A	0
VMVMVM	1	1	-1	-1	ZERO	N/A	0
VMVMVV	-1	-1	-1	-1	PLEAT	C	1
VMVVMV	-1	1	-1	-1	ZERO	N/A	0
VMVVMV	1	1	1	-1	ZERO	N/A	0
VMVVVM	1	-1	1	-1	ZERO	N/A	0
VMVVVV	-1	1	1	-1	ZERO	N/A	0
VMVVVV	-1	-1	1	-1	CONSTANT_RHO_MINUS	C	4k-3
VVMMMM	1	1	1	1	ZERO	N/A	0
VVMMMV	1	-1	1	1	ZERO	N/A	0
VVMMVM	-1	1	1	1	ZERO	N/A	0
VVMMVV	-1	-1	1	1	CONSTANT_RHO_PLUS	CR	6k-5
VVMVMM	1	-1	-1	1	ZERO	N/A	0
VVMVMV	1	1	-1	1	ZERO	N/A	0
VVMVVM	-1	-1	-1	1	PLEAT	CR	1
VVMVVV	-1	1	-1	1	ZERO	N/A	0
VVVMMM	-1	1	-1	1	ZERO	N/A	0
VVVMMV	-1	-1	-1	1	PLEAT	C	1
VVVMMV	1	1	-1	1	ZERO	N/A	0
VVVVVV	1	-1	-1	1	ZERO	N/A	0
VVVVVV	-1	-1	1	1	CONSTANT_RHO_PLUS	C	6k-5
VVVVMV	-1	1	1	1	ZERO	N/A	0
VVVVVV	1	-1	1	1	ZERO	N/A	0
VVVVVV	1	1	1	1	ZERO	N/A	0

Canonical representatives

The four canonical structural representatives are shown below. This table is a compact presentation of the already-proved structural mechanisms; it does not add a proof premise.

Word	ρ	τ	Mechanism	Compatibility	Multiplicity
MMMVVM	+1	-1	pleat Hamiltonian precedence + full Butterfly audit	N/A	one exact labelled state
MVMVMV	-1	-1	pleat row-monotone/reverse-row chain + full Butterfly audit	N/A	one exact labelled state
MMMMVV	+1	+1	row-local pivots + global merge	a=1 OR a=m OR a=b	one labelled state per compatible ordered pair
MVMMMM	-1	+1	LIFO horizontal reconstruction + pivot blocks	B_R(a)=B_R(b)	one labelled state per compatible ordered pair

Boundary physical projections

Projection	δ_0	Exact state	formal representatives
MMMM	1	EMPTY	4
MMMV	-1	A1,B1,B2,A2	4
MMVM	-1	B2,A2,A1,B1	4
MMVV	1	EMPTY	4
MVMM	-1	A1,A2,B2,B1	4
MVMV	1	EMPTY	4
MVVM	1	EMPTY	4
MVVV	-1	A2,A1,B1,B2	4
VMMM	-1	B2,B1,A1,A2	4
VMMV	1	EMPTY	4
VMVM	1	EMPTY	4
VMVV	-1	B1,B2,A2,A1	4
VVMM	1	EMPTY	4
VVMV	-1	B1,A1,A2,B2	4
VVVM	-1	A2,B2,B1,A1	4
VVVV	1	EMPTY	4